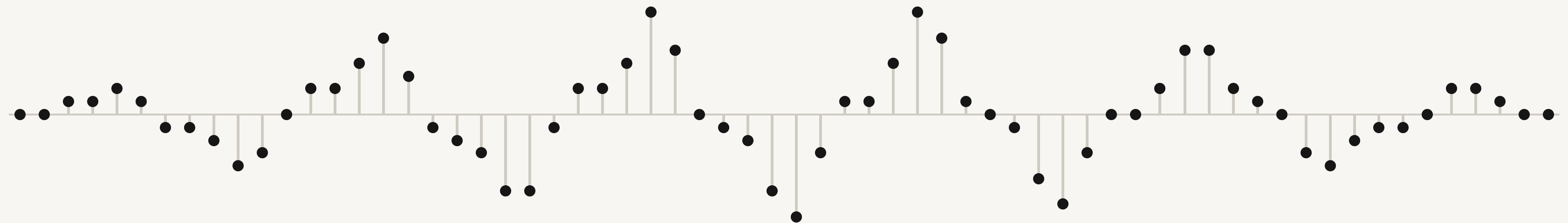


Beyond the Dropdown

Audio codecs for game audio, from ADPCM to neural

Collin Schupman, Audio Programmer

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About me

Collin Schupman

Audio Programmer

Riot Games	Audio programmer, unannounced R&D title
High Moon Studios	Audio engineer, Call of Duty
Output	Audio plugins and desktop software
Future Cities Lab	Interactive exhibition systems
Yeti	iOS development
Gas Powered Games	Audio engineer

Shipped

Call of Duty: Modern Warfare 4, Black Ops 6 and Black Ops 7

Output Arcade, Portal, Thermal and Movement

Works with

Custom game engines and audio engines, Unreal, Wwise, C++, real-time DSP, JUCE, audio plugins, iOS

Why this talk

I've worked on proprietary engines and commercial ones. Both have the same knowledge gap: how compressed audio actually works.

Audio is rendering. It is as deep a topic as graphics, and the two share most of their problems.

Graphics	Audio
Texture compression	Audio codecs
Frame budget	Buffer deadline
VRAM, texture streaming	RAM, audio streaming
Mips and LOD	Bitrate tiers, voice limits
GPU-decoded formats	Hardware decoders (XMA)

What a codec setting affects

Most projects pick a codec once, from a dropdown. That one setting decides:

File size	Install footprint and patch size
Streaming	Disk and I/O bandwidth under load
Memory and CPU	Resident RAM, and decode cost per voice
Platform compliance	Format requirements and certification
Sound quality	What players actually hear

01

The basics

Audio rendering and how codecs work

The audio render loop

1,024

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48,000

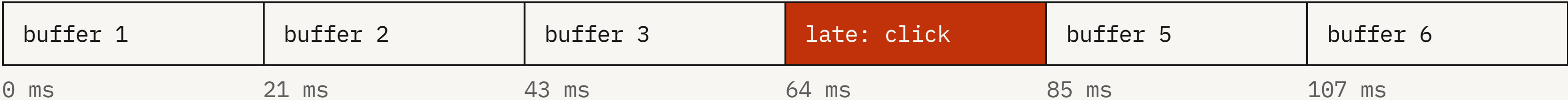
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≈ 21 ms

samples in a typical buffer

samples per second

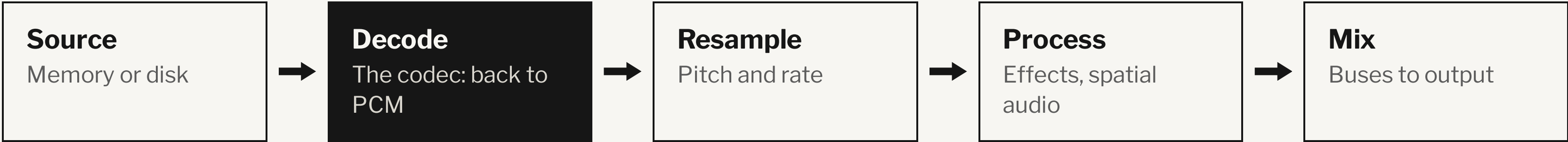
to render each buffer



Graphics	16.7 ms per frame at 60 fps. Miss it and you drop a frame.
Audio	About 21 ms per buffer. Miss it and players hear a click or dropout.

Decoding has to fit in that window, along with everything else in the audio frame.

The path of one voice



Where the source lives

Decoded in memory	PCM sits in RAM. No decode cost, most memory.
Compressed in memory	Decoded as it plays. Less RAM, CPU per voice.
Streamed from disk	Read in chunks, then decoded. Least RAM; adds I/O and latency.

Decode runs once per active voice, every buffer.

The cost of uncompressed audio

48,000

×

16

×

2

=

1,536 kbps

samples per second

bits per sample

channels

uncompressed stereo

Content	Size as PCM
One minute, stereo	11.5 MB
One hour, stereo	691 MB
One hour, 5.1	2.07 GB

Then multiply by every variation, language and platform.

Three kinds of codec

Kind	How it saves bits	What it costs	Examples
Waveform	Predicts each sample and stores only the error	Fixed ratio, some noise	ADPCM
Perceptual	Drops what masking hides from the ear	Decode CPU and latency	Vorbis, Opus, AAC
Lossless	Removes redundancy only; every sample comes back	Modest savings	FLAC

Every codec pays for its savings in CPU, latency or quality.

Inside a perceptual codec



Masking	A loud sound hides quieter ones near it in frequency and time. Those get fewer bits.
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Encode	At build time. Slow and thorough, and it runs once.
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Decode	At runtime. The same steps in reverse, per voice, every buffer.
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Terms on the settings panel

Bitrate	Bits per second of audio, in kbps. The main dial for size and quality.
CBR and VBR	Constant spends the same bits throughout. Variable spends more on complex passages.
Compression ratio	PCM size over encoded size. ADPCM is about 4:1; perceptual codecs often pass 10:1.
Frame size	Samples per coded block. Sets latency and how finely playback can seek.
Encoder delay	Extra samples added at the start. Left untrimmed, they break seamless loops.
Seek table	An index into the stream, so playback can start mid-file.

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The problems codecs solve

Samples, memory, voice chat, streaming and hardware

Sample-based audio

Early consoles synthesized sound on a chip. Once games played back recordings, the sample data had to fit in very little memory.

Year	Hardware	Audio memory	Compression
1990	Super Nintendo	64 KB audio RAM, 8 voices	BRR, an ADPCM variant, about 3.5:1
1994	PlayStation	512 KB sound RAM, 24 voices	4-bit ADPCM; XA-ADPCM streams from CD
1996	Nintendo 64	Samples share the cartridge ROM	ADPCM
2000	PlayStation 2	2 MB sound RAM, 48 voices	4-bit ADPCM
2005	Xbox 360	Shared system RAM	XMA, decoded in hardware

ADPCM won that era because the hardware could decode it for almost nothing.

Memory and disk

Minutes of 48 kHz stereo that fit in 100 MB



Dialogue	10 hours of mono VO is 3.5 GB as PCM. At 48 kbps Opus it is about 216 MB.
Languages, platforms	Each one multiplies the total, on disk and in every patch.

Voice chat

Live speech has to go both ways, in real time, over a connection that drops packets.

768 kbps

mono PCM at 48 kHz

→

≈ 24 kbps

Opus speech, 32× smaller

Low bitrate	Upstream bandwidth is shared with the game's own network traffic.
Low delay	Frames of 20 ms or less, so conversation stays natural.
Packet loss	The decoder conceals missing frames. Forward error correction fills gaps.
Lineage	Speex (2003), then SILK and CELT, merged into Opus (2012).

Codecs in streaming

In a game, streaming reads a file from disk at a bitrate you chose. Over a network, the codec has to adapt to the connection.

Where	Codecs	What the codec has to handle
Music services	Vorbis, AAC, Opus; FLAC or ALAC for lossless tiers	Quality per bit, and a decoder on every device
Video on demand	AAC family, including xHE-AAC; Dolby Digital Plus	Adaptive bitrate: several encodes, switched per segment
Live streams	AAC	Real-time encoding on the streamer's machine
Voice and calls	Opus	Low delay, and recovery from lost packets
Cloud gaming	Usually Opus	A delay budget shared with video and input

Three codecs carry most of it: Opus, Vorbis and AAC.

Hardware

Several codecs exist because of a specific hardware limit.

Dedicated decoders	Sound chips and DSPs decode ADPCM, XMA or AAC, so the CPU doesn't have to.
Battery	Hardware decode draws less power, which matters on handhelds and phones.
Storage media	Optical discs seek slowly, so music and VO stream as compressed data. Cartridge and download sizes cap the total.
Output links	Optical S/PDIF carries only stereo PCM, and Bluetooth is narrower still. Both need a codec on the way out.

A short history

1973

ADPCM

Predictive coding from Bell Labs. Still in every engine.

1993

MP3

Perceptual coding goes mainstream.

1997

AAC

MP3's successor, later decoded in phone hardware.

2002

Vorbis

Royalty-free. Becomes the game default.

2005

XMA

Xbox 360's format, decoded in hardware.

2011

ATRAC9

Sony's low-latency codec, from PS Vita on.

2012

Opus

IETF standard. Speech and music in one codec.

Waveform coding first, then perceptual coding, then platform-native codecs, then one flexible open codec.

ADPCM and MP3

Why it survives

ADPCM

- Fixed 4:1 against 16-bit PCM
- Near-zero decode CPU
- Sample-accurate seeks and loops
- Still right for short, busy SFX

Why it faded

MP3

- Encoder delay and padding add silence
- Seamless loops break
- Patent licensing until 2017
- Vorbis did the job for free

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The modern toolbox

Opus, Vorbis, AAC, XMA, ATRAC9, and how to choose

Codecs in use today

Codec	Good for	Watch out for	Wwise / Unreal
ADPCM	Short, frequent SFX	Fixed ~4:1; hiss on bright sounds	Both
Vorbis	Music, ambience	Weaker than Opus at low rates	Wwise
Opus	VO, music, low bitrates	More decode CPU than ADPCM	Both
Bink / RAD	Unreal's default	Unreal only	Unreal
AAC	Apple targets	Licensing; loop padding	Wwise (Apple)
XMA	Xbox, hardware decode	Xbox only	Wwise; UE Platform Specific
ATRAC9	PlayStation, cheap decode	PlayStation only	Wwise; UE Platform Specific

Support varies by version and platform. Check your build.

Vorbis and Opus

The incumbent

Vorbis

- Royalty-free; 1.0 released in 2002
- Long-time default in Wwise and many engines
- Mature encoders, predictable decode
- Falls behind at low bitrates

The challenger

Opus

- IETF standard, RFC 6716 (2012)
- SILK for speech, CELT for music
- 6 to 510 kbps; frames from 2.5 to 60 ms
- Best quality per bit at low rates

Default new streamed content to Opus. Keep Vorbis where a pipeline already depends on it.

Platform codecs

Codec	Platform	What you get	What you give up
XMA	Xbox	Hardware decode; CPU freed for more voices	Xbox only
ATRAC9	PlayStation	Low latency and low decode cost	PlayStation only
AAC	Apple, Android	Hardware decoders on most phones	Licensing; loop padding

Each platform gets its own encode. Budget build time and QA for it.

Codecs you didn't choose



Bluetooth headsets	SBC, AAC, aptX, LDAC, and LC3 on LE Audio. Adds latency too.
TVs and soundbars	Bitstream output encodes the mix live as Dolby Digital or DTS.
Capture, streams, cloud	The final mix is encoded again, usually as AAC or Opus.

Lossy generations stack, so leave headroom in the in-game encode.

**Choose a codec per asset
class, not per project.**

A decision order

Ask in this order. Stop at the first yes.

1	Short, frequent, latency-critical?	ADPCM, or PCM if memory allows
2	Dialogue or voice chat?	Opus, at speech bitrates
3	Long and streamed from disk?	Opus or Vorbis
4	CPU-bound at peak voice count?	Platform codec: XMA, ATRAC9, AAC

Then listen on target hardware, at real mix levels, before locking bitrates.

What to measure

Decode CPU	Profile at worst-case voice count, not one sound in isolation.
Memory vs. streaming	Resident assets want cheap decode. Streams want ratio and seeking.
Loops and seeking	Check that sample-accurate loop points survive encoding.
Bitrate per class	Set per asset class, then listen on target speakers and headphones.
Platform rules	Check each platform's format requirements.
Tools and licensing	Confirm encoder support in the pipeline, and any license terms.

04

The frontier

Apple's APAC and neural audio codecs

Apple's APAC

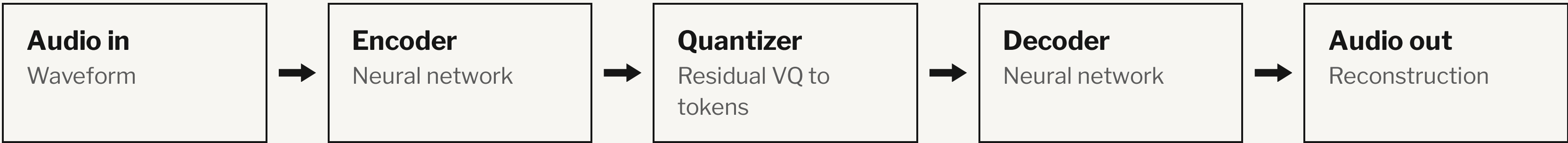
- Apple Positional Audio Codec. In the OS since iOS 13, formally defined with iOS 18.
- Carries channel-based and scene-based (Ambisonics) audio.
- Delivery format for Apple Spatial Audio Format (ASAF), shown at WWDC25.
- Required for Apple Immersive Video. Reported up to 768 kbps.
- Streams over HLS, with loudness and DRC metadata embedded.

For game audio

- Useful for linear content: cinematics, trailers, immersive video.
- Apple platforms only, encoded with Apple's OS tools.
- A delivery codec. Your engine still renders the interactive mix.

Neural codecs

A network learns the transform and the perceptual model from data.



Codec	From	Year	Notes
SoundStream	Google	2021	Streamable and causal; about 3 to 18 kbps
EnCodec	Meta	2022	General audio; about 1.5 to 24 kbps
Lyra	Google	2021–22	Speech only; v2 is built on SoundStream

Language models in codecs

Neural codecs turn audio into tokens, and a language model can predict tokens. That is used in two directions.

Approach	Example	What it does
LM as entropy model	EnCodec (2022)	A small Transformer predicts tokens; up to 40% fewer bits
LM fills in detail	LMCodec (2023)	Send coarse tokens; a Transformer generates the fine ones
Codec as LLM tokenizer	Mimi, in Moshi (2024)	12.5 frames/s, about 1.1 kbps; replies in about 200 ms
Ultra-low-rate speech	ZipCodec (2026)	0.8 kbps streaming speech from an 842M-parameter model
Promise	Speech near 1 kbps, and tokens an AI character can generate directly.	
Catch	The decoder generates plausible detail, so it can change a word or a voice. Most are trained on speech only.	

Barriers in games

Cost per voice	One model inference per stream, and a game mixes dozens of voices.
Latency	Many models need lookahead. Streaming variants trade quality for speed.
Training data	Mostly speech and music. Sharp SFX transients are unfamiliar.
Determinism	Float math on GPUs and NPUs can differ across platforms.
Model size	Weights ship with the game and compete for memory.
Tooling	Little middleware support today, and no standard bitstream.

Where neural audio shows up first

Now	ML inside classic codecs. Opus 1.5 (2024) added neural packet-loss recovery and decoder enhancement.
Next	Low-bitrate speech. Research codecs reach 20 ms latency at roughly 20× real time on one CPU.
Speculative	Tokens in place of waveforms. Generative pipelines that store and stream codec tokens, decoded on device NPUs.

Expect it in voice before music or SFX.

Takeaways

- 1 Codec choice is a budget decision: CPU, memory, disk and quality at once.
 - 2 Decide per asset class. ADPCM for short SFX, Opus for VO and streams, platform codecs where hardware helps.
 - 3 On the frontier, APAC is for spatial delivery, and neural codecs will reach voice first.
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Thank you

Questions?

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